

Aniket Deshpande

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EDUCATION

University of Illinois, Urbana-Champaign

2023–2027

B.S. in Physics, *Specialization in Physics & Computation*

Urbana, IL

Thesis (in progress): *Effective Many-Body Structure in Masked Language Models*

RESEARCH INTERESTS

Mechanistic Interpretability · Deep Learning Theory · Tensor Networks · Quantum Simulation

RESEARCH EXPERIENCE

Cambridge Boston Alignment Initiative

Fall 2026

Research Fellow

Cambridge, MA

- Developing theories of deep learning and optimization using tree tensor network models.

Lawrence Berkeley National Laboratory

May–Aug. 2026

Student Researcher; Mentor: Prof. Roel Van Beeumen

Berkeley, CA

- Owned the formulation, implementation, and validation of a matrix-free randomized range finder for moving the orthogonality column in isometric tensor networks, using random matrix-product-state probes and a bounded carried factorization.
- Built dense-oracle tests for sampling, projection, reconstruction, local and global isometry, and full-range state preservation; every measured defect was below 10^{-10} .
- On ten paired controls at heights four through six, reduced estimated floating-point work by 1.4 – $1.9\times$ and state infidelity by 3 – $22\times$ relative to the Moses Move; the research implementation remained 5 – $7\times$ slower in wall time.

Supervised Program for Alignment Research

Feb. 2026–Present

Research Fellow

Remote

- Developing theories of the statistical physics of feature learning.
- Co-developed TempBench’s hidden-Markov synthetic framework, extending ground-truth sparse-feature recovery from independent token firings to temporally correlated denoising and feature-coupling tasks.
- Led the headline backtracking case study: designed the locked 61-question MATH-500 cohort, steering and re-sampling protocol, LLM-judge evaluation, and grouped sparse-probe analysis; temporal crosscoders improved detection by roughly 40% and causal inducement by roughly 15% relative to a conventional TopK sparse autoencoder.

Vector Institute for Artificial Intelligence

Feb.–May 2026

Research Intern; PI: Prof. Alán Aspuru-Guzik

Toronto, ON

- Led the project end to end: designed the Fermionic Shadow Regressor architecture and experimental program and wrote the manuscript, with input from collaborators and the principal investigator.
- Built a Hartree–Fock-conditioned harmonic model that predicts 120 degree-two Majorana observables across molecular geometry and time, together with a line-spectrum screen and a fixed-budget time-resolution bound.
- Across four eight-qubit molecular benchmarks, outperformed matched Gaussian-process and Fourier-feature surrogates on the harder scans; finite-shot LiH training matched the infinite-shot reference at 1,000 Born samples per matchgate label.

Laboratory for Parallel Numerical Algorithms

Sep. 2024–Present

Undergraduate Researcher; PI: Prof. Edgar Solomonik

Urbana, IL

- Contributing to an annealed importance sampling algorithm with Glauber dynamics for approximating tensor network contractions, with applications to quantum simulation.
- Showing that alternating Mahalanobis distance minimization for CP decomposition performs maximum likelihood estimation under a Kronecker-structured noise covariance model.

Illinois Mathematics Lab

Aug.–Dec. 2024

Undergraduate Researcher; PI: Marius Junge

Urbana, IL

- Co-developed an explicit quantum circuit for simulating birth–death graph channels using controlled equality checks, rotations, permutation gates, ancillas, and a trace-and-reset operation.
- Established an $O(n)$ upper bound and $\Omega(\sqrt{n})$ lower bound for simulation length in the project’s resource model, alongside $O(n)$ local-gate and ancilla-space bounds.

PUBLICATIONS & PREPRINTS

Decoder-Preserving Sparse Autoencoders: Which Readouts Survive Sparse Compression? A. Deshpande. <i>arXiv:2607.17425</i>	2026
Fermionic Shadow Regression: Amortized Learning of Molecular Electron Dynamics A. Deshpande, K. Panicker, L. Mantilla Calderón, M. Bagherimehrab, A. Aspuru-Guzik. <i>4th International Workshop on Quantum Machine Learning at IEEE QCE 2026</i>	2026
Feature-Resolved Attention D. Manning-Coe, N. Aljaafari, J. Stephenson, K. Kapre, H. Jain, A. Deshpande. <i>Mechanistic Interpretability at ICML 2026</i>	2026
Crosscoding Through Time: Sparse Feature Discovery Across Sequence Positions D. Manning-Coe, H. Xuanyuan, A. Deshpande, A. Shportko, W. Fei. <i>Mechanistic Interpretability at ICML 2026</i>	2026

TALKS & POSTERS

Fermionic Shadow Regression: Amortized Learning of Molecular Electron Dynamics POSTER <i>Quantum Techniques in Machine Learning (QTML 2026), Stellenbosch, South Africa.</i>	Dec. 2026
POSTER <i>Molecular Machine Learning Conference (MoML), MIT, Cambridge, MA.</i>	Oct. 2026
TALK <i>4th International Workshop on Quantum Machine Learning at IEEE Quantum Week, Toronto, ON.</i>	Sept. 2026
Decoder-Preserving Sparse Autoencoders: Which Readouts Survive Sparse Compression? POSTER <i>New England Mechanistic Interpretability Workshop, Boston, MA.</i>	Aug. 2026
Approximating Tensor Contractions with Annealed Importance Sampling POSTER <i>QSim 2025, New York, NY.</i>	Aug. 2025
Quantum Circuit Volume for Graph Models POSTER <i>Illinois Mathematics Lab Open House, Urbana, IL.</i>	Dec. 2024

INDUSTRY EXPERIENCE

Space Dynamics Laboratory <i>Ionospheric Analyst Intern</i>	May–Aug. 2024 <i>Albuquerque, NM</i>
• Developed a Python data pipeline that collected more than 600 NICT ionograms per hour. • Implemented and benchmarked image-denoising filters in Python and Julia using PSNR, MSE, and SSIM, and investigated convolutional architectures for automatic ionogram scaling.	

SELECTED TRAINING

Independent Study in Information Geometry with Prof. Matthew Singh	Aug.–Dec. 2025
QSim Summer School — NSF RQS, hosted at IBM, New York, NY	Aug. 2025
Uncertainty Quantification & Machine Learning for Physical Systems — IMSI, University of Chicago	May 2025

SERVICE & AFFILIATIONS

Membership Director, SIAM at the University of Illinois, Urbana-Champaign	May 2025–Present
Member, Society for Industrial and Applied Mathematics	May 2025–Present

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Julia, Mathematica
Libraries: NumPy, SciPy, Pandas, Matplotlib, PyTorch, SymPy
Tools: Git, L^AT_EX, Conda, shell scripting, Jupyter